

Question 1

Total Marks: 5

Solution

Color the squares of the figure like a chessboard, using black and white alternately. Since every pair of horizontally or vertically adjacent squares has opposite colors, every domino covers exactly one black square and one white square.

The figure contains **42 squares**, so there are **21 black squares and 21 white squares**.

Construct a bipartite graph as follows:

- The left side L consists of the 21 black squares.
- The right side R consists of the 21 white squares.
- There is an edge between two vertices if and only if the corresponding squares are horizontally or vertically adjacent.

A domino corresponds to an edge of this graph. Therefore, a tiling of the entire figure by dominoes is equivalent to a **perfect matching** in this bipartite graph.

We now apply **Hall's theorem**.

Number the rows from top to bottom and the columns from left to right. Consider the following 11 black squares:

$$S = \{(1, 3), (2, 2), (2, 4), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (5, 1), (5, 3), (6, 2)\}.$$

Thus,

$$|S| = 11.$$

The white squares adjacent to at least one square in

S are

$$N(S) = \{(1, 4), (2, 3), (3, 2), (3, 4), (4, 1), (4, 3), (4, 5), (5, 2), (5, 4), (6, 3)\}.$$

Hence,

$$|N(S)| = 10.$$

Therefore,



$$|N(S)| = 10 < 11 = |S|.$$

This violates Hall's condition, which requires

$$|N(S)| \geq |S|$$

for every subset

S of the left side.

Thus, the bipartite graph **does not have a perfect matching**. Since a perfect matching exists if and only if the figure can be tiled by

1×2 or 2×1 dominoes, we conclude that



the given figure cannot be tiled by dominoes.

Marks Distribution — 5 Marks

- **1 Mark:** Correctly color/model the 42 squares as a bipartite graph with 21 vertices on each side.
- **1 Mark:** Explain why a domino tiling is equivalent to a perfect matching in the bipartite graph.
- **1 Mark:** Correctly identify a subset S of 11 vertices on one side.
- **1 Mark:** Correctly identify its neighborhood and show that

$$|N(S)| = 10 < 11 = |S|.$$
- **1 Mark:** Apply Hall's theorem and conclude that no perfect matching, and hence no domino tiling, exists.

Total: 5 Marks

Argument: "If we subtract the given figure

F from a perfectly tilable bounding rectangle A , and the remaining shape B cannot be tiled, then F must be non-tilable."

Counterexample-

Let

A be a standard 4×4 square grid (which is tilable).

Let F be the set of squares $(1, 2)$, $(2, 2)$, $(2, 1)$, and $(3, 1)$.

This shape is simply two vertical dominoes, so F is perfectly tilable. If we remove F from A , the top-left square $(1, 1)$ loses both of its neighbors and becomes completely isolated. This makes the remainder B non-tilable.

Question 2

Total Marks: 5

Proof of Correctness (3 Marks)

- **0.5 Mark:** Establishing the base case for the proof by induction, noting that if $x = 0$, the algorithm correctly returns a product of 0.
[$x = 1$ is also given marks]
- **1.25 Mark:** Providing the inductive step or recursive identity for even numbers, demonstrating that $x \cdot y = \lfloor x / 2 \rfloor \cdot (y + y)$ if x is even.
- **1.25 Mark:** Providing the inductive step or recursive identity for odd numbers, demonstrating that $x \cdot y = \lfloor x / 2 \rfloor \cdot (y + y) + y$ if x is odd.

Alternative Rubric: Proof of Correctness (Binary/Bitwise Approach - 3 Marks)

Proof of Correctness (3 Marks)

- **1 Mark (Parity to LSB):** Correctly identifying that the `if x is odd` condition is mathematically equivalent to checking whether the current Least Significant Bit (LSB) of x 's binary representation is 1.

- **1 Mark (Operations to Shifts):** Correctly mapping the arithmetic operations to bitwise shifts. The student must state that $x \leftarrow \lfloor x / 2 \rfloor$ acts as a right-shift (discarding the processed bit of x) **AND** that $y \leftarrow y + y$ acts as a left-shift (scaling y by powers of 2 for the respective bit position).
- **1 Mark (Formal Mathematical Synthesis):** Rigorously synthesizing the above points using formal mathematical notation (e.g., defining $x = \sum_i x_i 2^i$) and proving that the loop accumulates $\sum_i x_i (y \cdot 2^i)$, which equals $x \cdot y$.

Indexed notation, with loop variables separated from inputs.

Inputs

x, y with $x \geq 0$; bit length $n = \lfloor \log_2 x \rfloor + 1$; expansion $x = \sum_{i=0}^{n-1} x_i 2^i$ with $x_i \in \{0, 1\}$;

and $x^{(k)}, y^{(k)}, prod^{(k)}$ for the variable values after k iterations. Writing " x becomes $x / 2$ " without this separation makes an invariant unstateable, and is the most common reason an otherwise good binary answer caps at 2.

The invariant stated as a proposition over

k .

For all $0 \leq k \leq n$:

$$x^{(k)} = \left\lfloor \frac{x}{2^k} \right\rfloor, \quad y^{(k)} = 2^k y, \quad prod^{(k)} = y \sum_{i=0}^{k-1} x_i 2^i.$$

Runtime Analysis (2 Marks)

- **1 Mark:** Correctly identifying that the `while` loop executes $O(\log x)$ times because x is halved in each iteration.
- **1 Mark:** Stating the correct overall time complexity based on a clearly stated assumption for the time complexity of the addition operation. Applies to either of the following cases:
 - * **Case 1:** Assuming addition takes $O(1)$ time, concluding the overall runtime is

$O(\log x)$.

* **Case 2:** Assuming addition takes $O(\log y)$ or $O(n)$ time, concluding the overall runtime is $O(\log x \cdot \log y)$ or $O(n \log x)$.

* *Note:* If $O(n \log x)$ is given and n is not explicitly defined as $\log y$, this is still accepted for full credit.

Deductions (Runtime Analysis):

- **-1 Mark:** If the assumption for the addition complexity is not stated and the final complexity is written directly.
- **-1 Mark:** If the student uses m (e.g., in an $O(mn)$ final answer) to represent $\log x$ but fails to explicitly define $m = \log x$ in their solution.

Question 3

Total Marks: 5

- **0 Marks (Fatal Errors):**
 - Award **0 marks** for the question if the student provides an $O(n^2)$ solution.
 - Award **0 marks** if the Divide and Conquer approach is incorrectly split into 4 terms (e.g., setting up a recurrence that calculates 4 subproblems instead of 2, which ultimately leads back to an $O(n^2)$ complexity).
- **2 Marks:** Correctly expressing the matrix-vector product in the split block form:

$$H_k v = \begin{bmatrix} H_{k-1} & H_{k-1} \\ H_{k-1} & -H_{k-1} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} H_{k-1}v_1 + H_{k-1}v_2 \\ H_{k-1}v_1 - H_{k-1}v_2 \end{bmatrix}$$

- *Deduction:* Deduct **0.5 marks** if the size of vectors v_1 and v_2 (length $n / 2$ or 2^{k-1}) is not explicitly mentioned.

- **1 Mark:** Providing a clear recursive explanation, specifically identifying the $n / 2$ split and stating the size of the resulting subproblems and mentioning the $O(n)$ work needed to merge.
- **1 Mark:** Correctly formulating the time complexity recurrence relation (e.g.,
$$T(n) = 2T(n / 2) + O(n).$$
- **1 Mark:** Correctly solving the recurrence to arrive at $O(n \log n)$ with a proper explanation (such as referencing the Master Theorem).
 - *Partial Credit:* Award only **0.25 marks** (instead of 1) if the final time complexity is stated directly without an explanation using the Master Theorem or without a clear recursive explanation.